



ST BENEDICT'S

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|-----------|----------------------------|
| SUBJECT   | PHYSICAL SCIENCE - PHYSICS |
|           | NOVEMBER EXAM – PAPER 1    |
| GRADE     | 11                         |
| EXAMINER  | Mr Hilder                  |
| MODERATOR | Mr Ntombela                |
| DATE      | 03 November 2025           |
| MARKS     | 135                        |
| DURATION  | 2 hours                    |

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|               |                            |                            |                            |          |                             |                             |                              |
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| LEARNER NAME: |                            |                            |                            |          |                             |                             |                              |
| KEY:          | D <input type="checkbox"/> | E <input type="checkbox"/> | F <input type="checkbox"/> | TEACHER: | ZN <input type="checkbox"/> | KH <input type="checkbox"/> | FMC <input type="checkbox"/> |

**PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY**

1. This question paper consists of 8 pages and a Data Sheet. Please check that your question paper is complete.
2. Read the questions carefully.
3. Answer **ALL** the questions on the question paper and hand it in at the end of the assessment.
4. Remember to write your NAME AND SURNAME in the space provided above AND tick the block indicating your KEY and TEACHER'S INITIAL.
5. Use the data and formulae sheet whenever necessary.
6. Show your working in all calculations.
7. Units need not be included in the working of calculations, but appropriate units should be shown in the answer.
8. Answers must be expressed in decimal format, not left as proper fractions.
9. Where appropriate, express answers to **TWO** decimal places.
10. It is in your own interest to write legibly and to present your work neatly.
11. Blank pages and blank graphs are included at the end of the paper. If you run out of space for a question, use this space. Clearly show the question number of your answer should you use this extra space.

| Question       | 1  | 2  | 3  | 4  | 5  | 6  | 7 | 8 | 9  | Total |
|----------------|----|----|----|----|----|----|---|---|----|-------|
| Mark           |    |    |    |    |    |    |   |   |    |       |
| Marker Initial |    |    |    |    |    |    |   |   |    |       |
| Moderated Mark |    |    |    |    |    |    |   |   |    |       |
| Question Total | 18 | 23 | 15 | 18 | 16 | 17 | 5 | 7 | 16 | 135   |

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| Comment:             |                      |
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| Learner's Signature: | Teacher's Signature: |

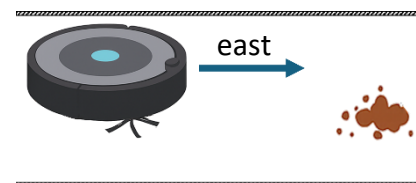
## QUESTION 1

18 MARKS

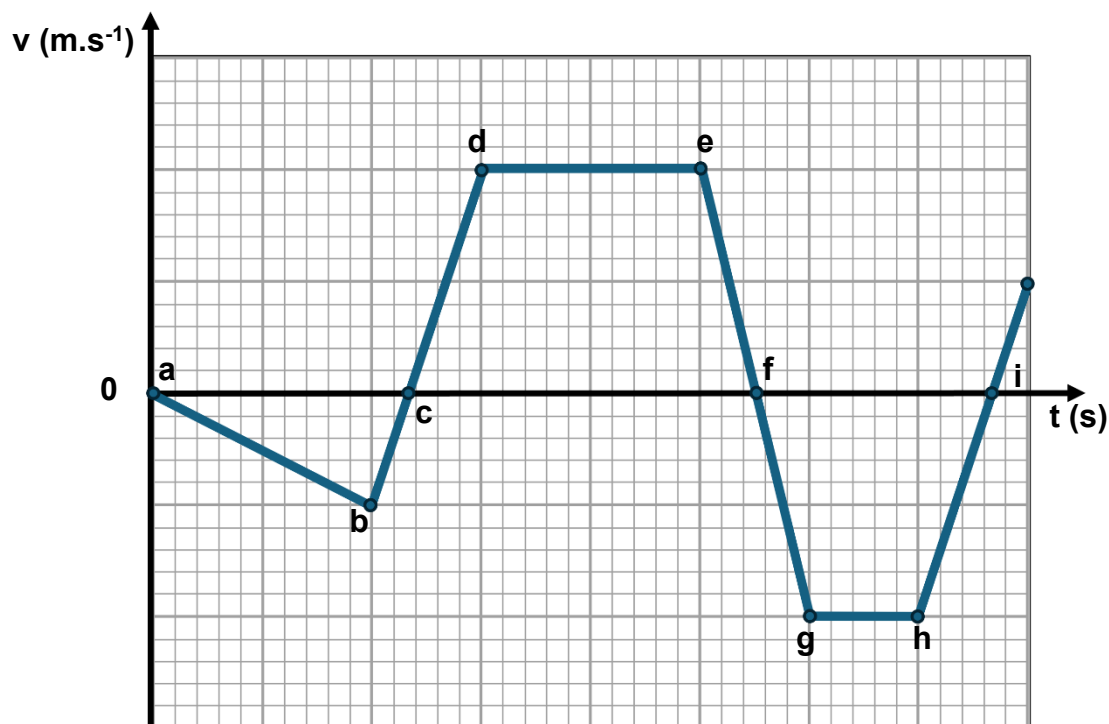
- 1.1 A Roomba (an automatic vacuum cleaner) moves in a straight line across the floor, attempting to clean a dirty spot.

Initially, the Roomba travels east for some distance from

**Point a**, it then changes direction and moves (in a straight line) toward its initial position. The Roomba continues moving variably for some time until **Point i**.



The graph below shows the velocity throughout the Roomba's entire motion.

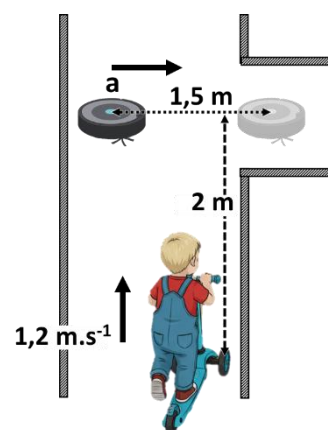


- 1.1.1 Define *velocity*. (2)
- 1.1.2 During which interval(s) was the Roomba moving with the greatest westward velocity? (2)
- 1.1.3 At which point(s) did the Roomba change direction? (2)
- 1.1.4 During which interval(s) was the magnitude of the Roomba's acceleration greatest? Briefly explain how you got to your answer. (3)

- 1.1.5 Draw a sketch graph to show the Roomba's acceleration with time over the period shown. On your graph, label points **a – i**. (4)



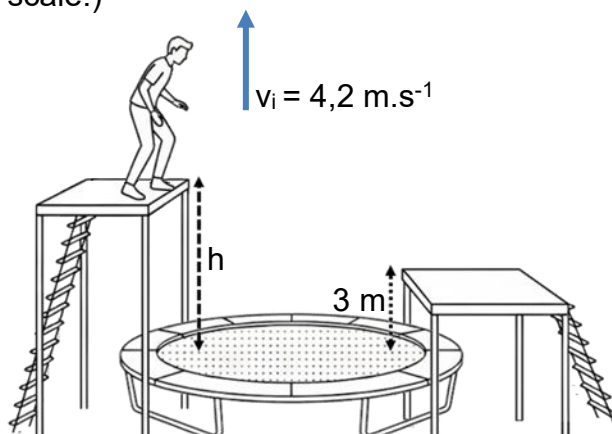
- 1.2 A child rides a push scooter down a passage at a constant speed of  $1,2 \text{ m} \cdot \text{s}^{-1}$ .  
The Roomba, initially at rest, is  $2 \text{ m}$  ahead in the child's path.  
As soon as it detects the child, the Roomba accelerates sideways across the  $1,5 \text{ m}$  wide hallway to avoid a collision.  
Calculate the minimum acceleration the Roomba needs to just clear the hallway before the scooter reaches it.



(5)

**QUESTION 2****23 MARKS**

A 70 kg person is on a high platform at a trampoline park. They jump upwards with an initial velocity of  $4,2 \text{ m.s}^{-1}$ . After experiencing free fall for 1,57 seconds, they land on the trampoline. (Diagram not drawn to scale.)



- 2.1 Define the term *displacement*. (2)
- 2.2 Calculate the height ( $h$ ) of the first platform above the trampoline. (3)
- 2.3 Calculate the person's maximum displacement relative to the trampoline during their initial jump. (4)
- 2.4 Show that the magnitude of the person's velocity just before landing on the trampoline is approximately  $11,19 \text{ m.s}^{-1}$ . (3)

The person lands on the trampoline, which stretches 0,8 m downward to bring them momentarily to rest. The trampoline then propels them upward to a new maximum height of 3,6 m above its surface, before they land on a second platform located 3,0 m above the trampoline.

- 2.5 Draw a labelled free-body diagram showing all forces acting on the person as the trampoline brings them to rest. (2)

- 2.6 Determine the average force exerted by the trampoline on the person as it brings him to rest over the downward stretch of 0,8 m. (4)

- 2.7 Sketch a vertical displacement–time graph of the person's motion from the jump off the first platform to landing on the second platform 3.0 m above the trampoline. Use the trampoline as the zero reference and label key points/values from Question 2

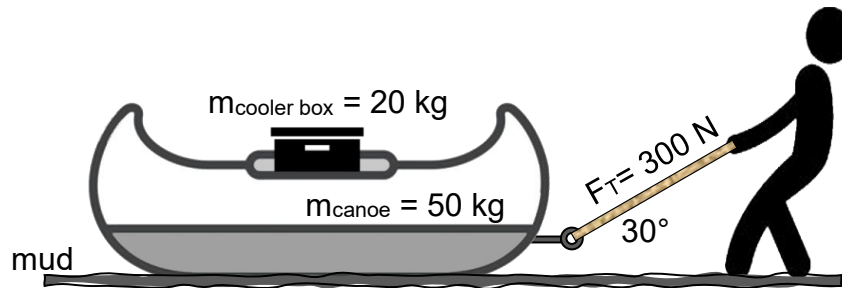
(5)



**QUESTION 3****15 MARKS**

A boy tries to pull his 50 kg fiberglass canoe, which contains a 20 kg cooler box, across a patch of flat, muddy ground. The total combined mass is 70 kg. The boy pulls on a rope with a constant force of 300 N ( $F_T$ ) at an angle of  $30^\circ$  to the horizontal.

The coefficient of static friction ( $\mu_s$ ) between the muddy ground and the canoe is 0,6.



- 3.1 Draw a labelled free-body diagram showing all the forces acting on the canoe as the boy pulls on it. (4)

- 3.2 Explain what is meant by the term *maximum static frictional force*. (2)

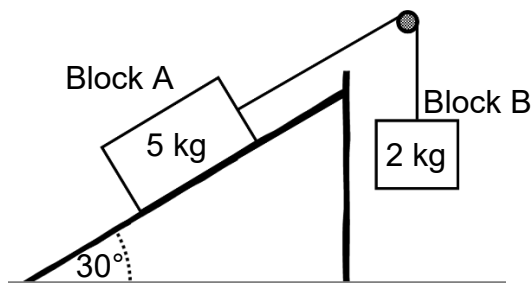
- 3.3 Calculate the normal force acting on the canoe. (3)

- 3.4 Determine, by means of a full calculation, that the boy will not be able to move the **fully loaded canoe (70 kg)**. (4)

- 3.5 The boy removes the cooler box, and is now able to move the canoe, using the same force being applied at the same angle.  
Provide an explanation as to why he is able to move the canoe. (2)

**QUESTION 4****18 MARKS**

**Block A** (mass: 5 kg) rests on a rough  $30^\circ$  incline. It is connected by a light, inextensible string that passes over a frictionless pulley to **block B** (mass 2 kg) which hangs vertically. The system has to be held stationary as when released, the blocks begin accelerating. While accelerating, 2N of friction acts on **block A**.



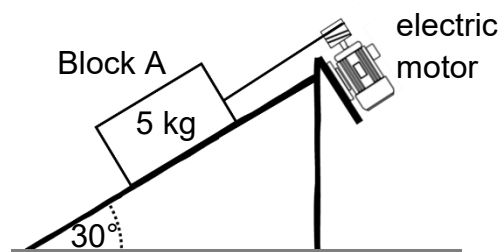
- 4.1 How does the tension in the string **compare** to the weight of block B while the system accelerates?  
Circle: **GREATER THAN, LESS THAN, or EQUAL TO.** (1)
- 4.2 Justify your answer to Question 4.1, refer to the forces on block B and Newton's Second Law. (2)
- 4.3 Calculate the tension in the string to validate the answer to 4.2. (4)
- 4.4 How does the magnitude of the system's acceleration **compare** to the acceleration on Earth due to Gravity?  
Circle: **GREATER THAN, LESS THAN, OR EQUAL TO.** (1)

When the block reaches the end of the slope it has gained 1,59 J of kinetic energy

4.5 State the *work-energy theorem*. (2)

4.6 Use ENERGY PRINCIPALS to determine how far block B travelled along the slope after being released. (3)

4.7 Block B is replaced by an electrical motor, the block is moved to the bottom of the slope. The perfectly efficient motor pulls the 5 kg block up the same slope at a constant velocity of  $0,5 \text{ m.s}^{-1}$ .

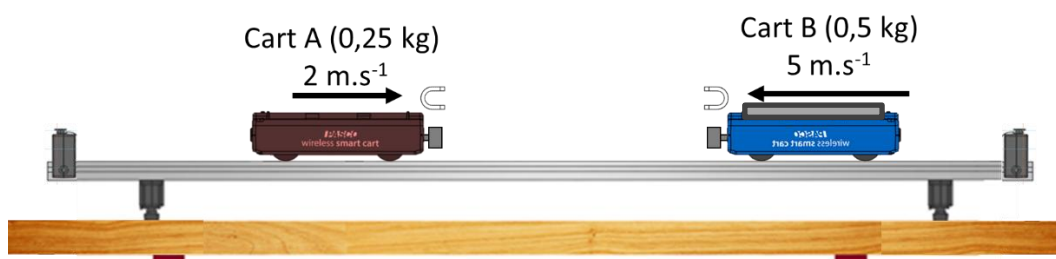


Determine the electrical power that must be **supplied to the motor** to pull the block up the slope. (5)

**QUESTION 5****16 MARKS**

In a physics experiment, two Pasco smart carts move on a long, frictionless track. Cart A has a mass of 0,25 kg and moves towards the right at a speed of  $2 \text{ m.s}^{-1}$ .

It is about to collide head-on with Cart B, which has a mass of 0,5 kg and is moving towards the left at a speed of  $5 \text{ m.s}^{-1}$ . Both carts have opposing magnets attached to the front that repel during the collision.



The carts undergo a perfectly **elastic** collision and do not stick together. The collision lasts for **0,05 seconds**.

- 5.1 State the *principle of conservation of linear momentum* in words. (2)
- 5.2 By applying the principles of conservation of both momentum and kinetic energy, calculate the final velocity (magnitude and direction) of both Cart A and Cart B immediately after the collision. (6)

- 5.3 Explain which cart, if either, experiences the greater force during the collision. (1)
- 5.4 Define *impulse*. (2)
- 5.5 Consider cart A moving after the collision. Calculate the magnitude of the force that acted on it during the collision, if they collision took 20 milliseconds. (3)
- 5.6 The magnetic collision attachments were used to improve the results of this experiment. Why would the magnetic attachments be used, rather than a rubber bumpers or springs? (2)

**QUESTION 6****17 MARKS**

- 6.1 An astronaut on a planet wants to determine the acceleration due to gravity. The astronaut has a number of different masses available and determines the weight of each mass. The following measurements were recorded by the astronaut:

| Mass (kg) | Weight (N) |
|-----------|------------|
| 0,1       | 0,30       |
| 0,2       | 0,79       |
| 0,3       | 1,05       |
| 0,5       | 1,90       |
| 0,6       | 2,18       |
| 0,7       | 2,70       |

- 6.1.1 Define *Gravitational Field*. (2)

- 6.1.2 Plot a graph of weight (on the y-axis) vs mass (on the x-axis) on the graph paper provided in the answer sheet. No Heading is necessary and the x-axis has been labelled for you. (4)

Graph showing relationship between weight and mass

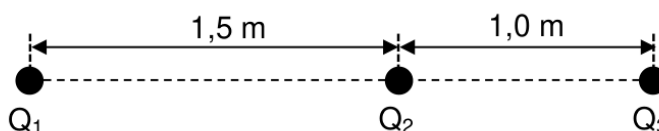


- 6.1.3 Determine the gradient of the graph and its unit. Then refer to the Table A below to determine which planet the astronaut was on. (4)

TABLE A

| Planet  | $g \text{ (m}\cdot\text{s}^{-2}\text{)}$ |
|---------|------------------------------------------|
| Venus   | 8,87                                     |
| Mars    | 3,71                                     |
| Jupiter | 23,12                                    |
| Pluto   | 0,58                                     |

- 6.2 In the diagram below,  $Q_1$ ,  $Q_2$  and  $Q_3$  are three-point charges placed along a straight line. All three charges are fixed in position. The distance between  $Q_1$  and  $Q_2$  is 1,5 m and the distance between  $Q_2$  and  $Q_3$  is 1,0 m as shown in the diagram below.



The magnitude of charges  $Q_1$  and  $Q_2$  are unknown. The charge on  $Q_1$  is positive. The charge on  $Q_3$  is  $+2 \mu\text{C}$  and  $Q_3$  experiences a net electrostatic force of 0,3 N to the left (towards  $Q_1$ )

- 6.2.1 Is the sign of charge  $Q_2$  positive or negative?  
Circle: **POSITIVE** or **NEGATIVE**. (1)
- 6.2.2 Explain your reasoning for QUESTION 6.2.1. (2)

**Charge  $Q_2$  is now removed. The magnitude of the electrostatic force experienced by  $Q_3$  due to  $Q_1$  is now 0,012 N.**

6.2.3 Calculate the magnitude of the unknown charge  $Q_1$  (4)

### QUESTION 7

**5 MARKS**

A teacher uses two different laser pointers in a demonstration.

**Laser A** emits **red light** (wavelength = 650 nm, power = 5 mW).

**Laser B** emits **green light** (wavelength = 530 nm, power = 5 mW).

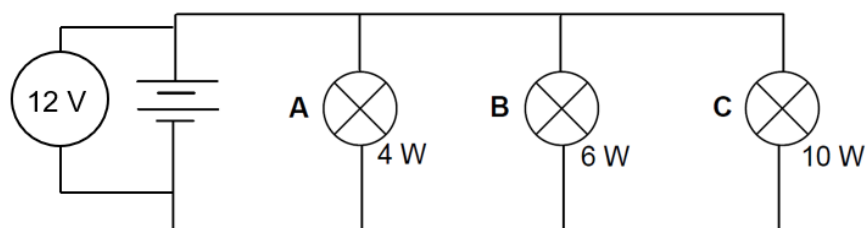
A student notes that the warning label on the green laser (Laser B) is much stricter, stating it is more dangerous to the human eye than the red laser (Laser A), even if both lasers have the same power output (intensity).

7.1 Explain why, between these two lasers with equal power, the green is more dangerous to the human eye. (2)

7.2 Determine the energy transferred by a single wave in green laser beams, provide the answer in electron Volts (eV) (3)

**QUESTION 8****7 MARKS**

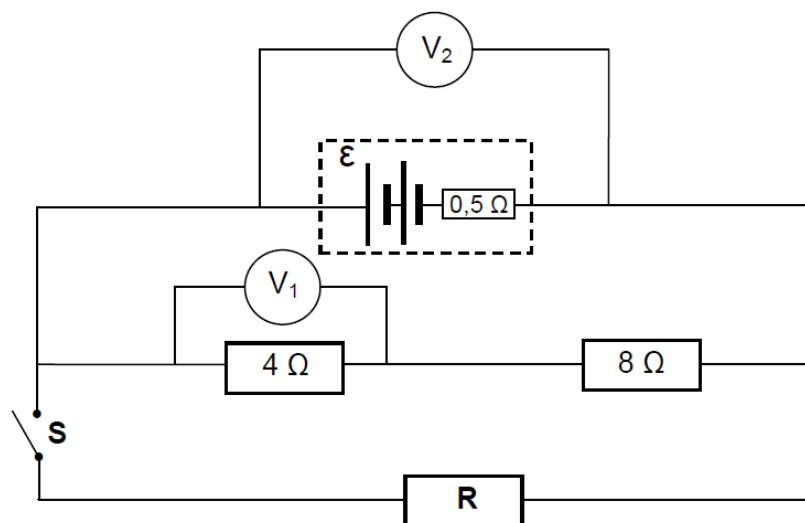
In the diagram below, three light bulbs, A, B and C, are connected in parallel to a 12 V source with a significant internal resistance. The bulbs are rated at 4 W, 6 W and 10 W respectively and are all at their maximum brightness. The voltmeter connected over the battery reads 12V.



- 8.1 Calculate the resistance of the 4 W bulb. (2)
- 8.2 How will the equivalent resistance of the circuit change if the 6 W bulb burns out?  
Circle: **INCREASES, DECREASES or NO CHANGE.** (1)
- 8.3 How will the power dissipated by the 10 W bulb change if the 6 W bulb burns out?  
Circle: **INCREASES, DECREASES or NO CHANGE.**  
Give a reason for the answer. (4)

**QUESTION 9****16 MARKS**

In the circuit below a battery of UNKNOWN emf and an internal resistance of  $0,5\ \Omega$  is connected to two resistors of  $4\ \Omega$  and  $8\ \Omega$  each, and a resistor  $R$  of unknown resistance. Ignore the resistance of the connecting wires.



- 9.1 The three external resistors are ohmic conductors. Explain the meaning of the term ohmic conductor. (2)

When switch  $S$  is OPEN, voltmeter  $V_1$  reads  $3,2\ \text{V}$ .  
Calculate the:

- 9.2 Current through the battery. (3)

- 9.3 Emf of the battery (4)

**When switch S is CLOSED, voltmeter  $V_2$  reads 8,8 V.**

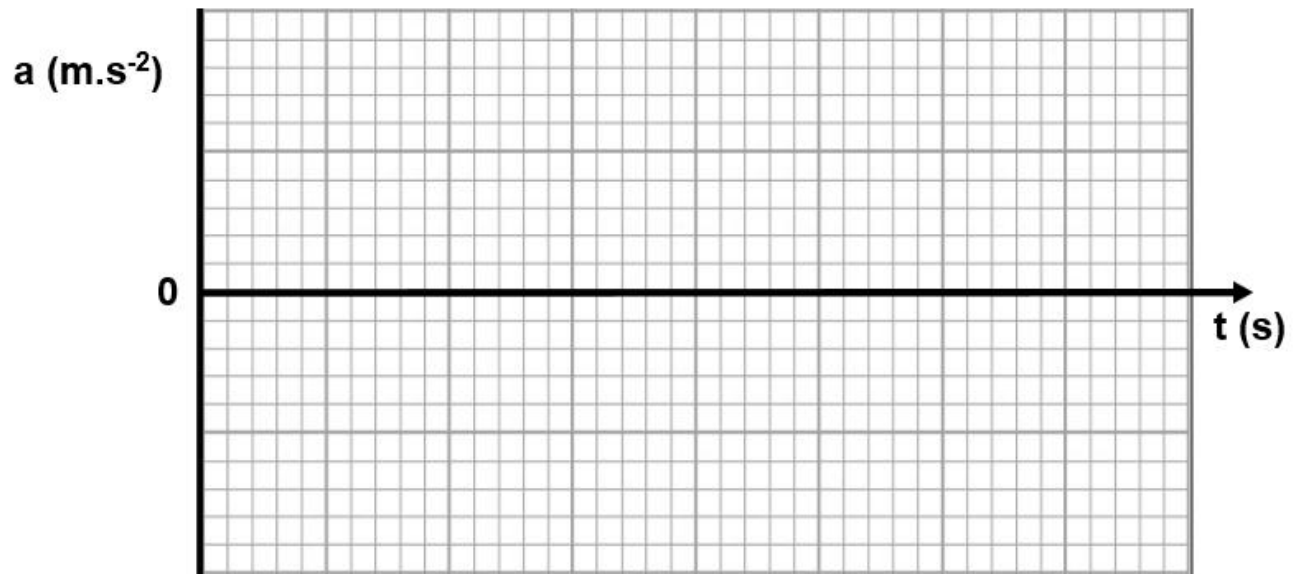
9.4 Calculate the resistance of resistor R. (5)

9.5 The battery becomes heated when voltmeter  $V_1$  is replaced by a connecting wire.  
Explain this observation. (2)

**END OF ASSESSMENT**

## ADDITIONAL GRAPHS

## QUESTION 1.1.5



## QUESTION 2.7



QUESTION 6.1.2

Graph showing relationship between weight and mass



ADDITIONAL SPACE

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